

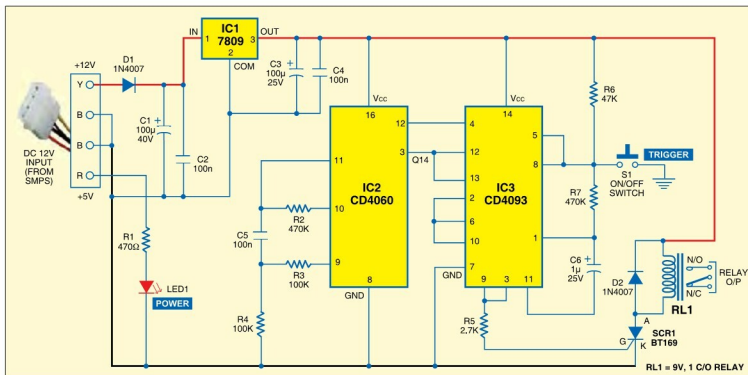


Fig. 1: PC timer circuit

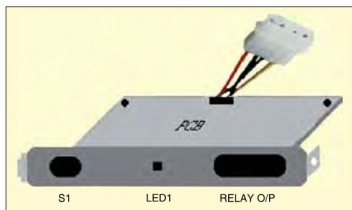


Fig. 2: Suggested enclosure of PC timer

To connect the input voltage, hook up the circuit's input supply points to the PC's internal power supply cabling. Any unused 4-way drive live power connector (+5V/0V/0V/+12V) of the ATX SMPS can be used for powering the circuit.

As shown in Fig. 1, the fixed three-terminal voltage regulator IC 7809 (IC1) provides 9V from the PC's SMPS to the circuit. Diode D1 protects against accidental polarity reversal. LED1 is a power on/off indicator.

The circuit is built around a 14-stage ripple-carry binary counter CD4060 (IC2) and a quad 2-input NAND Schmitt CD4093 (IC3).

The timer-controlled relay energises after the period determined by R-C time constant components R2 and C5. With component values as shown (R2=470 kilo-ohms and C5=100 nF),

the relay energises after 10 minutes and latches. The time-out period may be altered by changing the value of R2 and/or C5. You can use a preset, in series with R2 and pin 10 of IC2, to set the time.

To start the timer, simply press switch S1 momentarily. IC2 starts oscillating and after six seconds, the output of IC3 at pins 3 and 9 provides gate voltage to trigger SCR1. As a result, the relay energises and latches to activate the connected load. The relay remains latched until switch S1 is pressed again.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet as shown in Fig. 2. Provide a suitable connector on the PCB for connecting the power supply from the SMPS of your PC. ●

TIMER FOR MOSQUITO DESTROYER

■ D. MOHAN KUMAR

In electric-heating mosquito repellents, an electric vaporiser heats up a mat or liquid to release non-

degrading chemicals into the air and keep the mosquitoes away from the closed surroundings.

Here's a circuit that introduces a time gap in the operation of the va-

